

SIM7600G-H 4G HAT (B)

From Waveshare Wiki

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Feature

- Connected via Pogo PIN or MicroUSB connector.
 - Dedicated pogo pin for Raspberry Pi Zero/Zero W.
 - MicroUSB connector for other Raspberry Pi boards or PCs.
- Incorporates SIM7600G-H global band 4G module, compatible with 2G/3G/4G network with global support.
- USB HUB connector for other Raspberry Pi boards or PCs, providing USB extension and 4G network access.
- Supports dial-up, telephone calls, SMS, TCP, UDP, DTMF, HTTP, FTP, etc.
- Supports GPS, BeiDou, Glonass, and LBS base station positioning.
- SIM card slot, supports 1.8V/3V SIM card.
- Onboard audio jack and audio decoder for making the telephone calls.
- 2 x LED indicators, easy to monitor the operating status.
- Control via AT commands (3GPP TS 27.007, 27.005, and V.25TER command set).
- Supports SIM application toolkit: SAT Class 3, GSM 11.14 Release 99, USAT.
- Comes with online development resources and manuals (examples for Raspberry Pi).

SIM7600G-H 4G HAT (B)



(<https://www.waveshare.com/sim7600g-h-4g-hat-b.htm>)

Test

- To test this device, you should prepare the following components:

SIM7600G-H 4G HAT (B) × 1
LTE Antenna × 1
GPS Antenna × 1
USB-A to micro-B cable × 1
4G SIM card × 1
An earphone with a microphone × 1 (optional)

- Insert the activated 4G SIM card and the earphone (optional) into the HAT when the HAT powers off, and then connect it to the PC by the USB cable.
- Connect one end of the micro USB cable to the USB port of the PC, and the other end to the USB port of SIM7600X 4G HAT(B), the PWR indicator is on, and the NET indicator is not on.
- When the NET indicator starts to flash once every second, the module starts to work.
- External USB devices such as keyboards or mice are connected to the USB Type-A connector and can be recognized by the computer. (Optional)



(/wiki/File:SIM7600_4G_HAT(B)TOP_ang_BOTTOM.png)

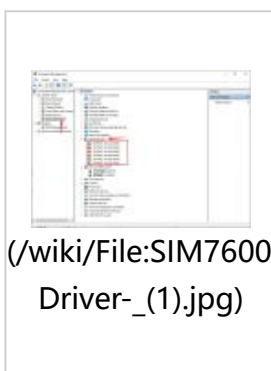
- Check the Device Manager of the PC, the several COM ports will be recognized:

端口 (COM 和 LPT)

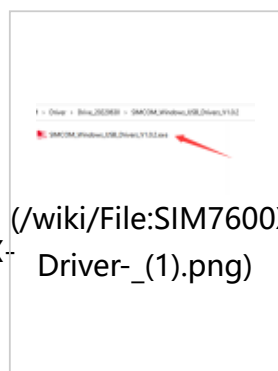
- SimTech HS-USB AT Port 9001 (COM19)
- SimTech HS-USB Audio 9001 (COM17)
- SimTech HS-USB Diagnostics 9001 (COM21)
- SimTech HS-USB NMEA 9001 (COM18)

(/wiki/File:Sim7600CE-3.jpg)

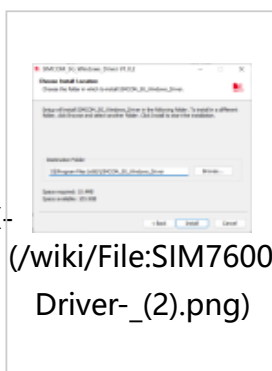
- If the COM ports are recognized with an exclamation symbol, you should install the SIM7600X Driver (https://files.waveshare.com/upload/2/24/SIMCOM_Windows_USB_Drivers_V1.0.2.zip) manually.
- Connect the 4G HAT to the Windows PC (Hereafter we use Windows 10 PC as an example)
- Unzip the driver file --> Double-click the exe driver file with the left mouse button --> Select the installation path --> NEXT --> Wait for the installation to complete --> Restart the computer --> Complete the driver installation.



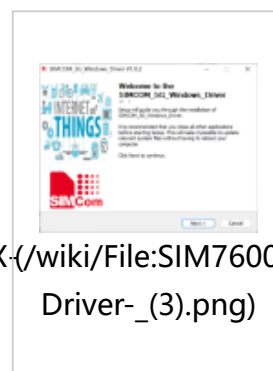
(/wiki/File:SIM7600X-Driver_(1).jpg)



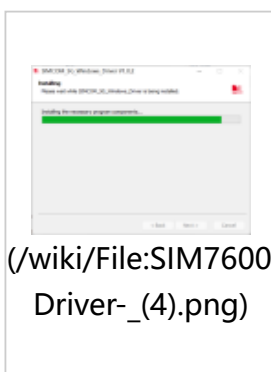
(/wiki/File:SIM7600X-Driver_(1).png)



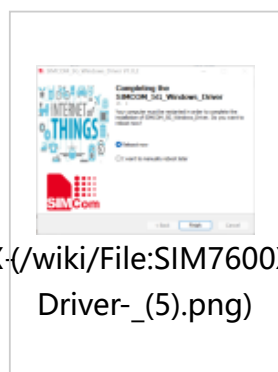
(/wiki/File:SIM7600X-Driver_(2).png)



(/wiki/File:SIM7600X-Driver_(3).png)

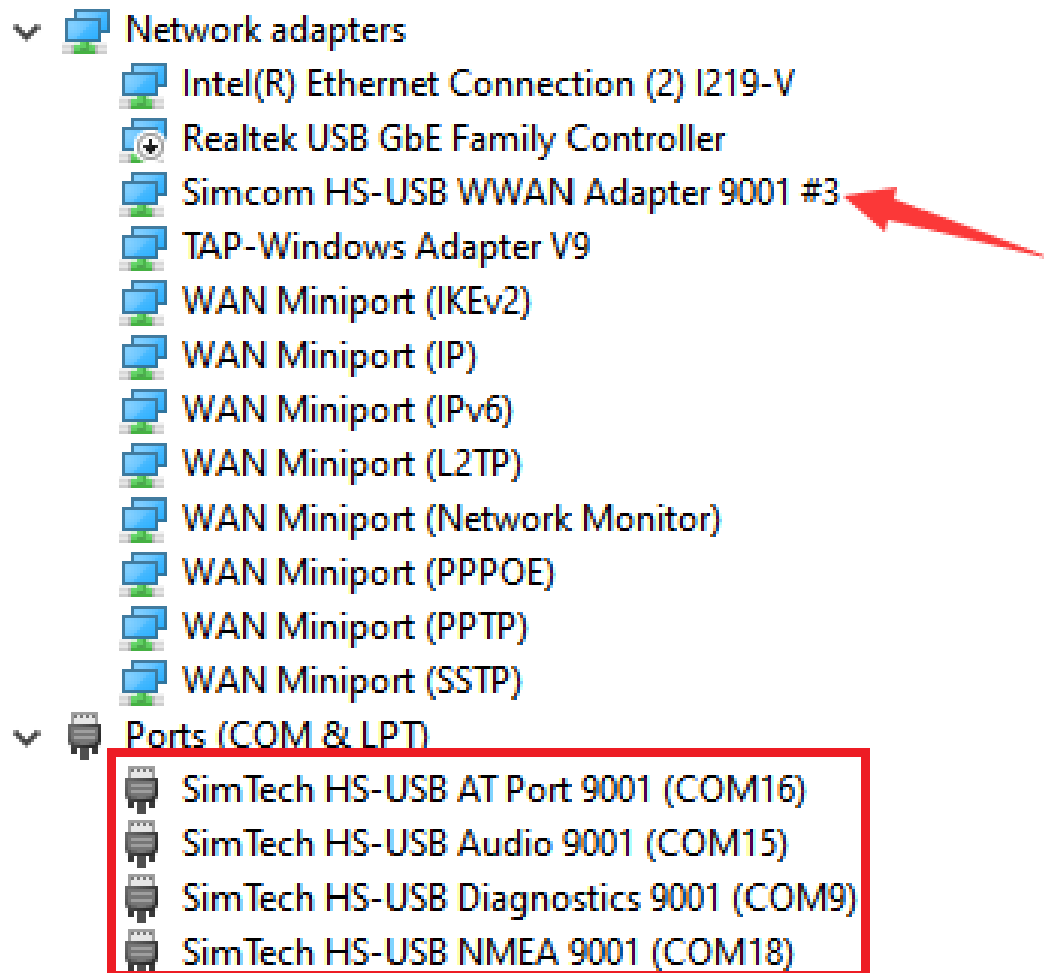


(/wiki/File:SIM7600X-Driver_(4).png)



(/wiki/File:SIM7600X-Driver_(5).png)

- After installation, all the devices should be recognized normally as below:



(/wiki/File:SIM7600X-Driver_(6).png)

NET Working Status

Reference of indicators

Table 17: NETLIGHT pin status

NETLIGHT pin status	Module status
Always On	Searching Network; Call Connect(include VOLTE,SRLTE)
200ms ON, 200ms OFF	Data Transmit; 4G registered;
800ms ON, 800ms OFF	2G/3G registered network
OFF	Power off ;Sleep

Note: NETLIGHT output low level as “OFF”, and high level as “ON”.

(/wiki/File:NETLIGHT.png)

AT Commands for Testing

AT Command	Description	Return
AT	AT Test Command	OK
ATE	ATE1: Echo mode on ATE0: Echo mode off	OK
AT+CGMI	Request manufacturer identification	OK
AT+CGMM	Request model identification	OK
AT+CGSN	Request product serial number identification	OK
AT+CSUB	Request product version	OK
AT+CGMR	Request firmware version	OK
AT+IPREX	Configure baud rate of model	+IPREX:OK
AT+CRESET	Reset model	OK
AT+CSQ	Query signal quality	OK
AT+CPIN?	Query SIM card status	+CPIN:RRADY
AT+COPS?	Query provider information	+COPS:OK
AT+CREG?	Query network registration status	+CREG: OK
AT+CPSI?	Query UE system information	
AT+CNMP	Select Network Mode: 2: Automatic 13: GSM only 38: LTE only 48: Any modes but LTE	OK

For more information about the AT command, you can refer to AT command manual.

SSCOM V5.13.1 Serial/Net data debugger, Author: Tintin, 2618058@qq.com

PORT COM_Settings Display Send_Data Multi_Strings Tools Help 联系作者 PCB打样

AT
OK
ATE1
OK
AT+CGMI
SIMCOM INCORPORATED
OK
AT+CGMM
SIMCOM_SIM7600G-H
OK
AT+CGSN
868822040351940
OK
AT+CSUB
+CSUB: B03V02
+CSUB: MDM9x07_LE20_LWG_22_V1.05_200828
OK
AT+IPREX?
+IPREX: 115200
AT+CSQ
+CSQ: 25, 99
OK
AT+CPIN?
+CPIN: READY
OK
AT+COPS?
+COPS: 0, 0, "CHINA MOBILE CMCC", 7
OK
AT+CREG?
+CREG: 0, 1
OK
AT+CPSI?
+CPSI: LTE, Online, 460-00, 0x27B6, 9438315, 409, EUTRAN-
BAND41, 41332, 5, 5, -75, -902, -628, 16
OK
AT+CNMP=2
OK

Send Multi Char | stm32/GD32 ISP | STC/IAP15 ISP |

<- Drag split ☐ RoundSend **help** Import order delay

HEX	Data (DbClick to Write notes)	Send (note)	+	-	ms
	General AT commands	十六进制数据串1	1		1000
	AT	AT test command	3		1000
	ATE1	Enable echo	2		1000
	AT+CGMI	Module manufacturers	0		1000
	AT+CGMM	Module model	0		1000
	AT+CGSN	Serial number	0		1000
	AT+CSUB	Module revision	0		1000
	AT+CGMR	Firmware revision	0		1000
	AT+IPREX?	Baud rate	0		1000
	AT+CRESET	Reset module	0		1000
	AT+CSQ	Signal quality	0		1000
	AT+CPIN?	SIM Card Status	0		1000
	AT+COPS?	Operator selection	0		1000
	AT+CREG?	Network registration	0		1000
	AT+CPSI?	UE system infor	0		1000
	AT+CNMP=2	auto mode selection	0		1000
	AT&F	factory settings	0		1000
	Phone Calls AT commands	18无注释	0		1000
	AT+CNRM	Phone number	0		1000
	AT+CSIDVC=1	handset output	0		1000
	AT+CLVL=?	check volume level	0		1000
	AT+CLVL=2	volume level setting	0		1000
		23None	0		1000
	ATD<phonenumber>:	make a call	0		1000
	AT+CHUP	Hang up call	0		1000
		26None	0		1000
	AT+CLIP=1	Phone ring	0		1000
	Serial output: RING	Phone ringing	0		1000
	ATA	Call answer	0		1000
	AT+CHUP	Hang up call	0		1000
		31None	0		1000
	SMS AT commands	32None	0		1000
	AT+CMGL="ALL"	List all messages	0		1000
	AT+CMGR=0	Read message	0		1000
	AT+CMGD=0	Delete message	0		1000
	AT+CNMI=2,1	message indications	0		1000
		37None	0		1000
	Sending English message	38None	0		1000
	AT+CMGF=1	Select message forma	0		1000
	AT+CSGS="GSM"	select TE character s	0		1000
	AT+CSMP=17,167,0,0	set text mode para	0		1000
	AT+CMGS="<phonenumber>"	Send message	0		1000
	Hello, Waveshare	message	0		1000

ClearData OpenFile SendFile Stop ClearSend OnTop ☒ English SaveConfig EXT

ConNum COM4 SinTech HS-USB AT For ☐ HEXShow ☒ SaveData ☐ ReceivedToFile ☐ SendHEX ☐ SendEvery: 1000 ms/Tim ☒ AddCrLf

☒ CloseCom ☒ More Settings ☐ Show Time and Packe OverTime: 20 ms No 1 BytesTo 末尾 Verify None

☐ RTS ☒ DTR BaudRate: 115200 simcommsg

为了更好地发展SSCOM软件
请您注册嘉立创结尾客户

【升级到SSCOM5.13.1】★嘉立创PCB打样SIM贴片服务 ★RT-Thread中国人的开源免费操作系统 ★81M远距离WiFi可自组网 ★新一代WiFi芯片兼容8266支持RT-Thread

www.daxia.com S:136 R:523 COM4 Opened 115200bps,8,1,None,None CTS=0 DSR=0 RLSD=1

(/wiki/File:SIM7600X_H_01.png)

Dial-up Networking

【Note】You must use a SIM card with GPRS networking enabled and haven't been out of service.

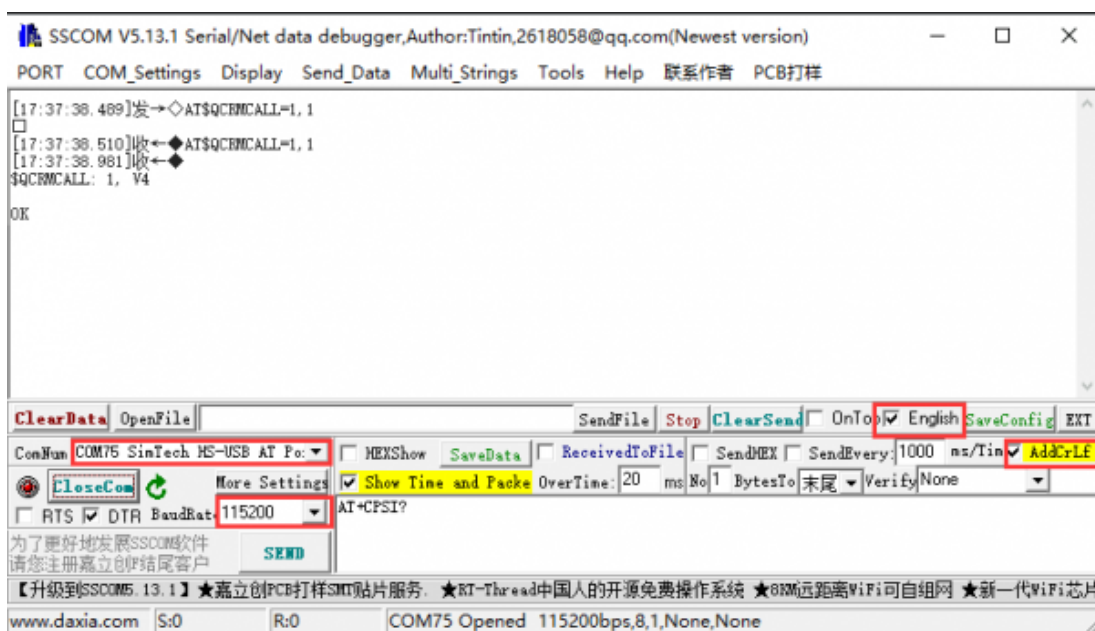
Dial-up Networking with Windows PC

- Connect the SIM7600G-H 4G HAT (B) with the 4G SIM card to your win10 PC. After installing the driver, the PC will automatically recognize the module.
- If a Windows PC still cannot access the Internet, you need to start NDIS dial-up manually.

Open the SIM7600 AT port and send the command:

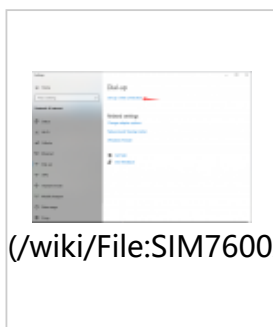
```
AT+QCRMCALL=1,1
```

Then press 'Enter'. At this time, NDIS dialing will take effect, and the PC can connect to the network.

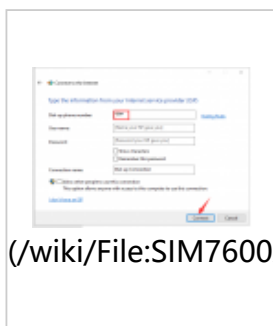
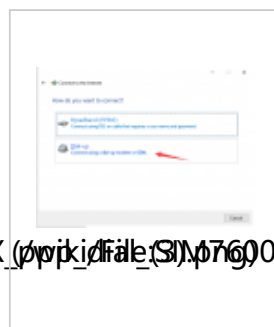
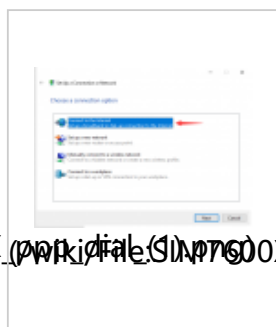


(/wiki/File:Sim8200_win_sendat1.png.png)

- If you still can't connect automatically, you can use PPP dial-up. The operation steps are as follows:



(/wiki/File:SIM7600X_ppp_dial(1).png) (/wiki/File:SIM7600X_ppp_dial(2).png) (/wiki/File:SIM7600X_ppp_dial(3).png) (/wiki/File:SIM7600X_ppp_dial(4).png)



(/wiki/File:SIM7600X_ppp_dial(5).png) (/wiki/File:SIM7600X_ppp_dial(6).png)



GPS Debugging

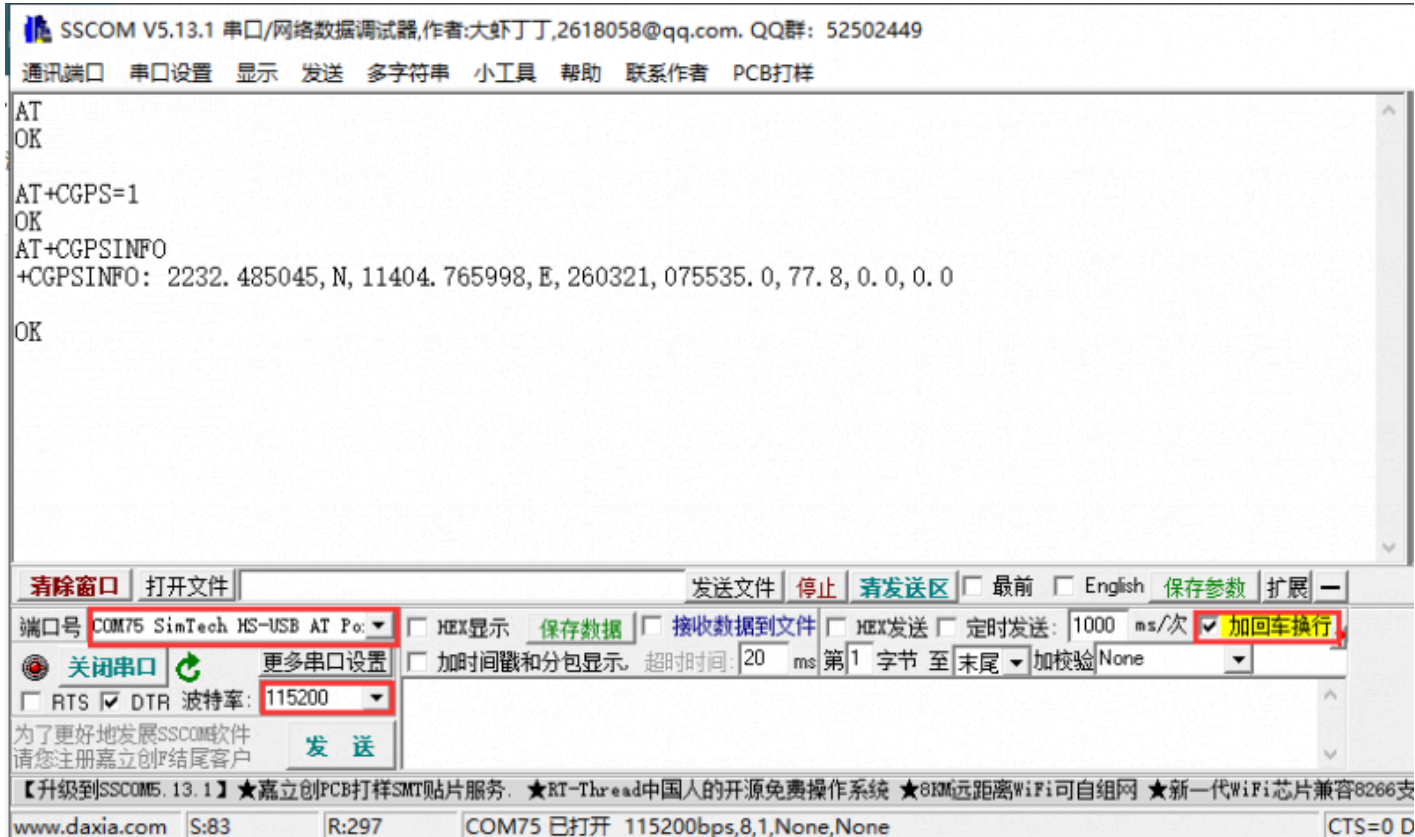
- Plug in the GPS antenna and place the 4G HAT in an open outdoor (note that it cannot be tested in rainy weather). It takes about 1 minute to receive the positioning signal after power-on.



(/wiki/File:SIM7600G-H-4G-HAT-B-details-9.jpg)

The test commands and screenshots are as follows:

```
AT+CGPS=1           //Open GPS
AT+CGPSINFO          //Print GPS information to the serial port
AT+CGPS=0           //Close GPS
```

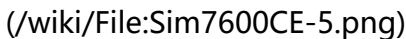


(/wiki/File:SIM7600G-H_4G_HAT(B)_GNSS.png)

Make Calls and Answer

- Insert the SIM card, connect the LTE antenna, and connect the USB interface of SIM7600E-H 4G HAT to the PC. Then press the PWRKEY to power on the board.
- Check whether the indicators blink correctly (PWR's and NET's flashes).
- Send AT commands as below:

Commands	Description	Return
AT+CNUM	Phone number (only supported by some SIM cards)	+CNUM OK
AT+CSDVC	AT+CSDVC=1: Handset output AT+CSDVC=3: Speaker output	OK
AT+CLVL=?	Check volume level	OK
AT+CLVL=2	Volume level set to 2	OK
ATD<phone_number>;	Make calls	OK
AT+CHUP	Hang up call	OK
AT+CLIP=1	Phone ring	OK
ATA	Answer call	OK



```
AT^PWRCTL=0,1,3      // Improve TDD noise effect
OK
```

- During the establishment of a voice call:

```
VOICE CALL:BEGIN      // Module call establishment and execution to improve call effect
AT+CECM=1             //Echo suppression processing
OK
AT+CECH=0x500         //Improve the volume effect of the mobile phone
OK
```

For more details, please check the file of "SIM7X00_Audio_Application_Note".

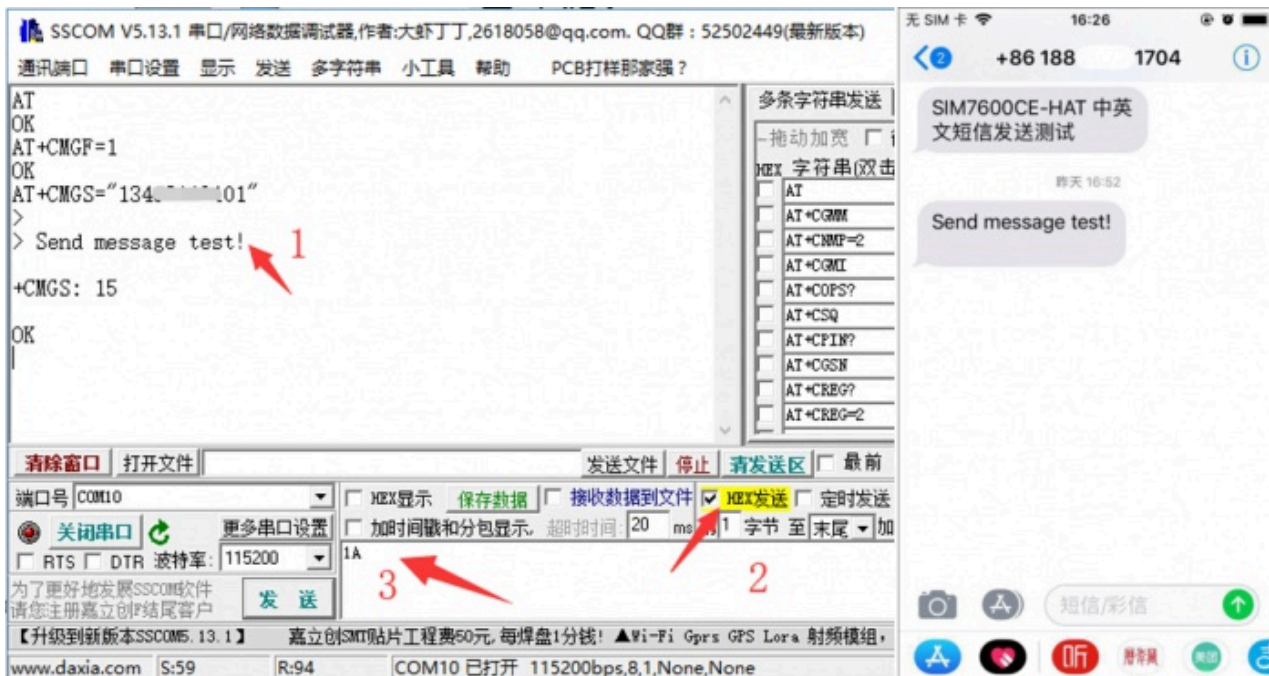
Send the Message

- 1. Plug the SIM card, connect the LTE antenna and connect the USB interface of the 4G HAT to PC, and power on the device.
- 2. Check whether the indicators blink correctly: the PWR indicator is always on, and the NET indicator is flashing.
- 3. Send AT commands as shown below:

```
AT+CMGF=1      //Set the message mode to TEXT;
```

- 4. Enter AT+CMGS="phone number" and then press <Enter> to set the recipient's mobile phone number, then this command will return: ">". And then you can enter the content to be sent, such as "Send message test!", and don't need to check 'AddCrLf'. After editing the message, send the information in HEX format, enter 1A in the send box, and then click SEND. After sending successfully, the module will return +CMGS: 15. You can see in the picture below:

【Note】 : 1A is the key value of "CTRL+Z", which is used to tell the module to perform the sending operation. You can also send 1B that is "ESC" to cancel the current operation.



(/wiki/File:Sim7600CE-6.jpg)

Receive the Message

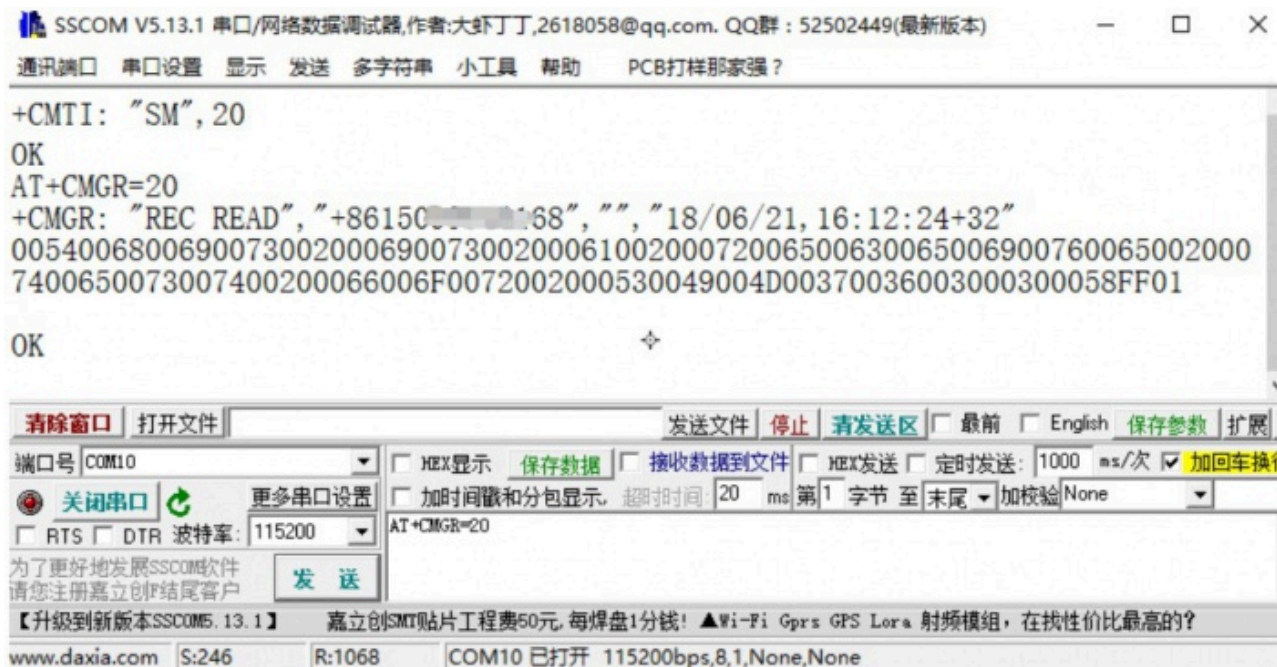
- 1. Send a message with your phone, such as "This is a receive test for SIM7600X!" to the module.
- 2. When receiving the information, the serial port will automatically report the information, "SM", 20, which means there are 20 messages in the SM, and the message just sent is the 20th message.
- 3. Read the message:

`AT+CMGR=20      //Read the 20th information (AT+CMGL="ALL" means to read all the information);`

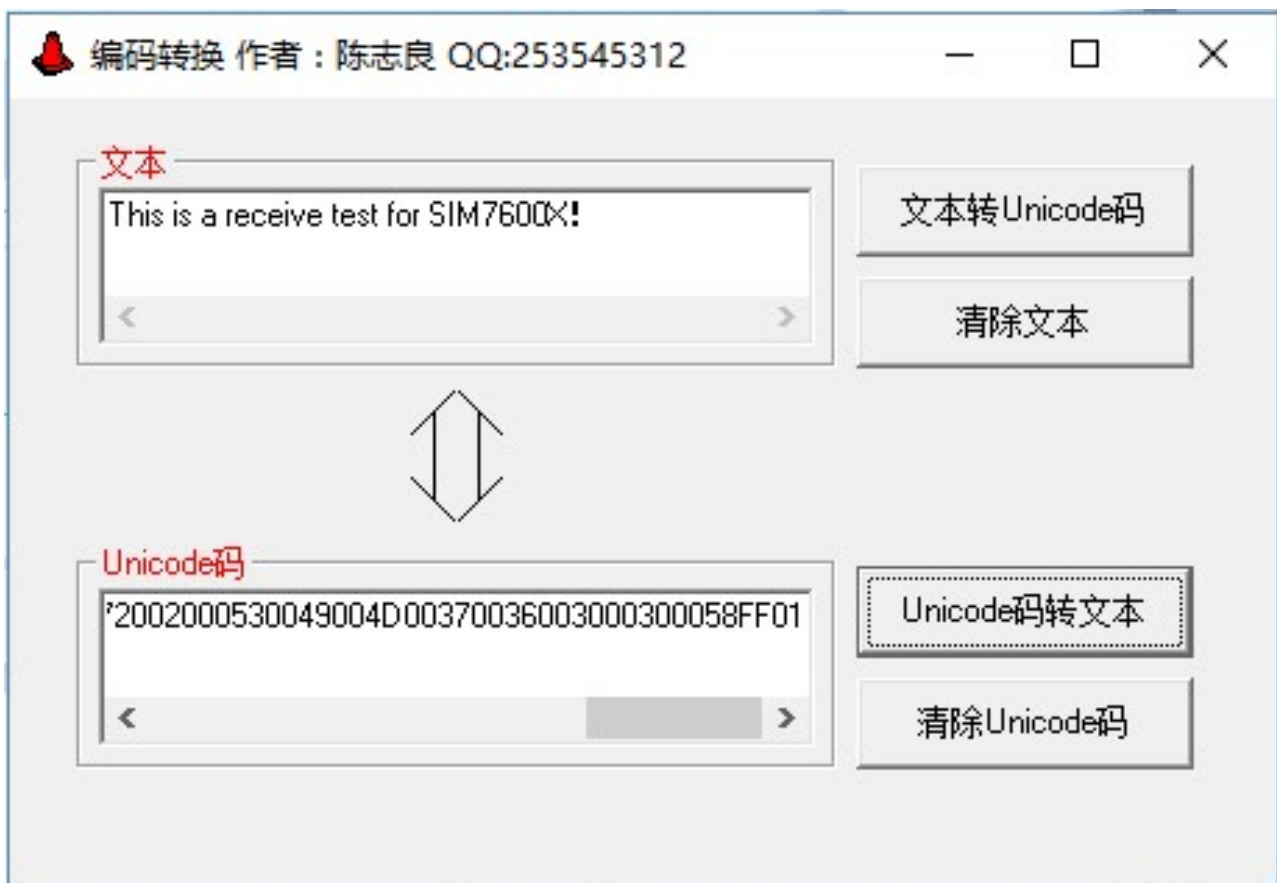
- 4. Delete the message:

`AT+CMGD=20`

- 5. Convert the displayed information into text through a code converter.



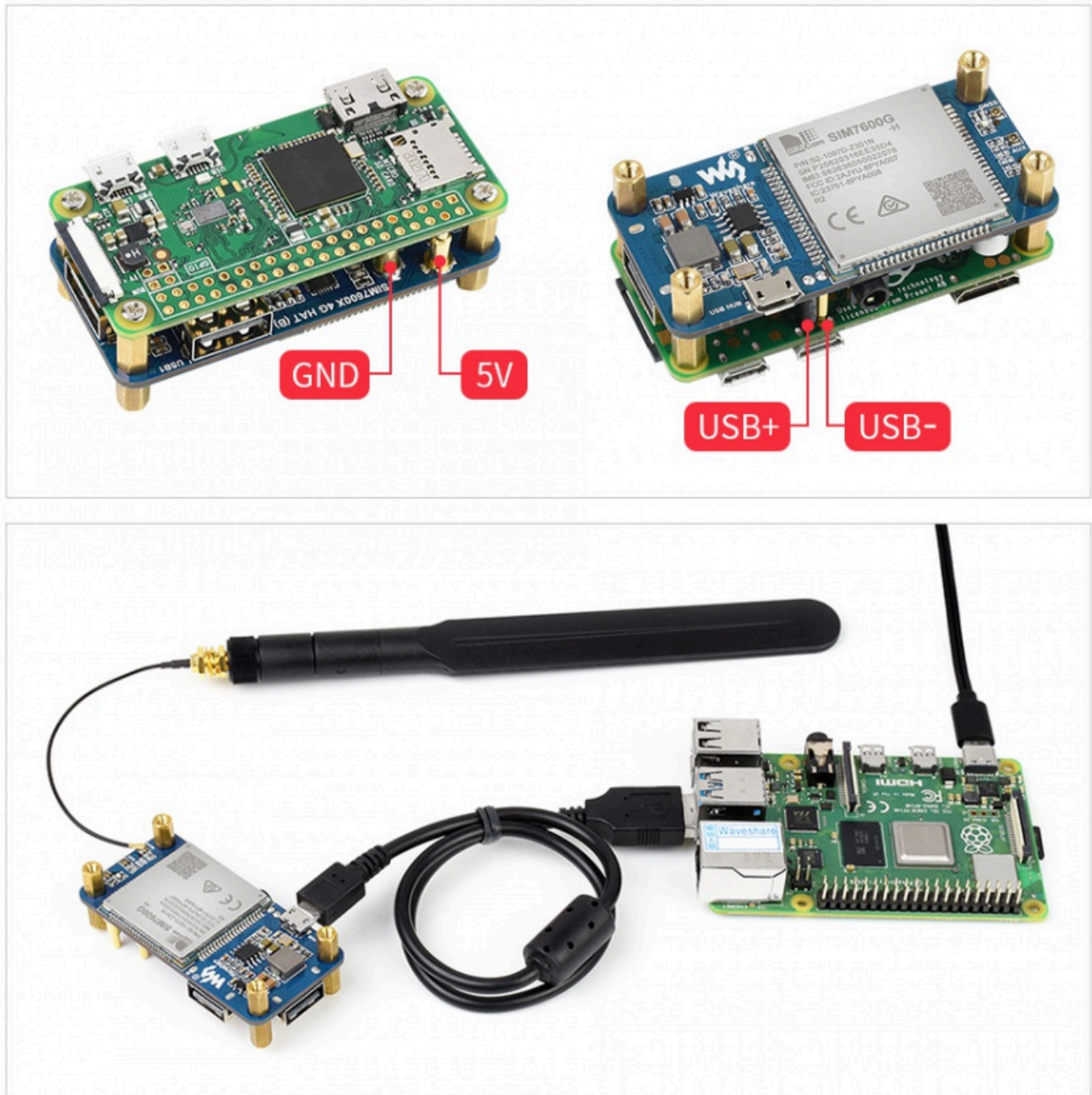
(/wiki/File:Sim7600CE-7.jpg)



(/wiki/File:Sim7600CE-8.jpg)

Dial-up Networking with Raspbian

- When using Raspberry Pi Zero/Zero W, you can connect it to the 4G HAT through the thimble, no additional USB cable is required;
- if it is used on other Raspberry Pi boards, you can connect the 4G HAT to the Raspberry Pi through a Micro USB cable. The connection is as follows:



(/wiki/File:SIM7600G-H-4G-HAT-B-details-3.jpg)

- And use the following method to Dial-up Networking
- Raspberry Pi networked via RNDIS (/wiki/Raspberry_Pi_networked_via_RNDIS)
- Raspberry Pi networked via NDIS (/wiki/Raspberry_Pi_networked_via_NDIS)
- SIM868 PPP Dial-up Networking (/wiki/SIM868_PPP_Dial-up_Networking)
- How to turn Raspberry Pi Zero W into a 3G 4G Router (https://www.waveshare.com/wiki/How_to_set_up_Raspberry_Pi_Zero_W_as_a_3G_4G_router)

Minicom Serial Debugging

- 1. Connect the HAT to your Pi through the USB cable.
- 2. Install minicom:

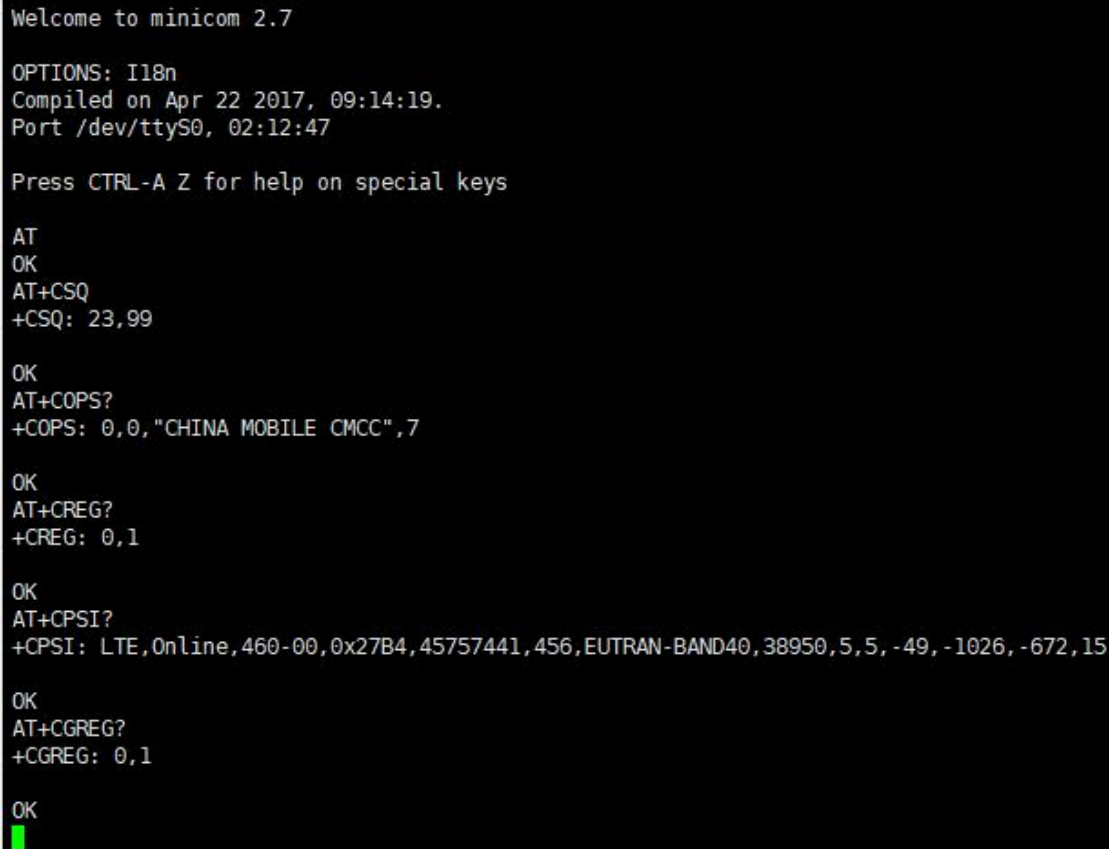
```
sudo apt-get install minicom
```

- 3. Execute the following command:

```
sudo minicom -D /dev/ttyUSB2
```

The default baud rate is 115200.

- 4. Take the AT synchronization test as an example, and send related commands, as shown in the picture below:



```
Welcome to minicom 2.7

OPTIONS: I18n
Compiled on Apr 22 2017, 09:14:19.
Port /dev/ttyS0, 02:12:47

Press CTRL-A Z for help on special keys

AT
OK
AT+CSQ
+CSQ: 23,99

OK
AT+COPS?
+COPS: 0,0,"CHINA MOBILE CMCC",7

OK
AT+CREG?
+CREG: 0,1

OK
AT+CPSI?
+CPSI: LTE,Online,460-00,0x27B4,45757441,456,EUTRAN-BAND40,38950,5,5,-49,-1026,-672,15

OK
AT+CGREG?
+CGREG: 0,1

OK
█
```

(/wiki/File:Sim7600CE-29.jpg)

Press Ctrl+A, then press Z, the minicom can enter the setting mode, and press X to exit.

Sample Demo

1. Insert the module and connect to the Raspberry Pi.
2. Download the sample demo to /home/pi/:

```
wget https://files.waveshare.com/upload/4/4e/SIM7600G-H-4G-HAT-B-Demo.zip
sudo apt-get install p7zip-full
7z x SIM7600G-H-4G-HAT-B-Demo.zip
sudo chmod 777 -R SIM7600G-H-4G-HAT-B-Demo
```

3. Respectively enter the corresponding directory, compile and run the demo. The related instructions are shown below (Take the PhoneCall.py demo as an example):

```
cd SIM7600G-H-4G-HAT-B-Demo/Raspberry/python/PhoneCall/  
sudo apt-get install python3-pip  
sudo pip3 install pyserial  
sudo pip3 install RPi.GPIO  
sudo python3 Phonecall.py
```

AT Command Demo

```
pi@raspberrypi:~/SIM7600G-H-4G-HAT-B-Demo/Raspberry/python/AT $ sudo python AT.py  
Please input the AT command:AT  
AT  
OK  
  
Please input the AT command:AT+CSQ  
AT+CSQ  
+CSQ: 28,99  
  
OK  
  
Please input the AT command:AT+CPSI?  
AT+CPSI?  
+CPSI: LTE,Online,460-01,0x2543,111925513,424,EUTRAN-BAND3,1650,5,5,-45,-832,-572,15  
  
OK
```

(/wiki/File:SIM7600G-H_4G_HAT_(B)_AT.png)

PHONECALL Demo

```
pi@raspberrypi:~/SIM7600G-H-4G-HAT-B-Demo/Raspberry/python/PhoneCall $ sudo python Phonecall.py  
AT+CSQ  
+CSQ: 31,99  
  
OK  
  
AT+CREG?  
+CREG: 0,1  
  
OK  
  
AT+CPSI?  
+CPSI: LTE,Online,460-01,0x2543,111925513,424,EUTRAN-BAND3,1650,5,5,-45,-800,-560,14  
  
OK  
  
ATD10010;  
OK  
  
Call disconnected  
  
VOICE CALL: BEGIN  
AT+CHUP  
VOICE CALL: END: 000018  
  
OK
```

(/wiki/File:SIM7600G-H_4G_HAT_(B)_AT02.png)

SMS Demo

[illegible]

(/wiki/File:SIM7600G-H_4G_HAT_(B)_AT03.png)

GPS Demo

```
pi@raspberrypi:~/SIM7600G-H-4G-HAT-B-Demo/Raspberry/python/GPS $ sudo python GPS.py
Start GPS session...
AT+CGPS=0
OK

AT+CGPS=1
OK

AT+CGPSINFO
+CGPSINFO: ,,,,,,,

OK

AT+CGPSINFO
+CGPSINFO: 2232.497849,N,11404.751098,E,220721,023940.0,-65.1,0.0,

OK

AT+CGPSINFO
+CGPSINFO: 2232.497283,N,11404.752159,E,220721,023942.0,-53.6,0.0,

OK

AT+CGPSINFO
+CGPSINFO: 2232.496108,N,11404.756955,E,220721,023945.0,-37.4,0.0,

OK

AT+CGPSINFO
+CGPSINFO: 2232.495649,N,11404.760647,E,220721,023947.0,-30.2,0.0,

OK

AT+CGPSINFO
+CGPSINFO: 2232.495408,N,11404.763606,E,220721,023950.0,-27.5,0.0,

OK
```

(/wiki/File:SIM7600G-H_4G_HAT_(B)_AT04.png)

TCP Demo

```
pi@raspberrypi:~/SIM7600G-H-4G-HAT-B-Demo/Raspberry/python/TCP $ sudo python TCP.py
AT+CSQ
+CSQ: 30,99

OK

AT+CREG?
+CREG: 0,1

OK

AT+CPSI?
+CPSI: LTE,Online,460-01,0x2543,111925513,424,EUTRAN-BAND3,1650,5,5,-53,-806,-517,14

OK

AT+CGREG?
+CGREG: 0,1

OK

AT+CGSOCKCONT=1,"IP","CMNET"
OK

AT+CSOCKSETPN=1
OK

AT+CIPMODE=0
OK

AT+NETOPEN
OK

+NETOPEN: 0

AT+IPADDR
+IPADDR: 10.37.53.144

OK

AT+CIPOPEN=0,"TCP","118.190.93.84",2317
OK

+CIPOPEN: 0,0

AT+CIPSEND=0,
>
Waveshare
OK

+CIPSEND: 0,9,9

RECV FROM:118.190.93.84:2317
+IPD9
Waveshare
send message successfully!
AT+CIPCLOSE=0
```

(/wiki/File:SIM7600G-H_4G_HAT_(B)_AT05.png)

FTP Demo

```

pi@raspberrypi:~/SIM7600G-H-4G-HAT-B-Demo/Raspberry/python/FTP $ sudo python FTP.py
AT+CFTPSSTART
OK

+CFTPSSTART: 0

AT+CFTPS SINGLEIP=1
OK

AT+CFTPS LOGIN="120.79.2.0",21,"aly","root",0
OK

+CFTPS LOGIN: 0

AT+CFTPS LIST="/"
OK

+CFTPS LIST: DATA,950
drwxr-xr-x  2 1001      www      4096 Jun 30 14:37 .
drwxr-xr-x  2 1001      www      4096 Jun 30 14:37 ..
-rwxr-xr-x  1 1001      www      1 May 20 00:46 .htaccess
-rw-r--r--  1 0         0        37 May 20 00:46 .user.ini
-rwxr-xr-x  1 1001      www      479 May 20 00:46 404.html
-rwxr-xr-x  1 1001      www      313 Jun 22 20:50 Jetson nano TTL.py
-rw-r--r--  1 1001      www      12 Jun  1 11:46 SIM7600.txt
-rw-r--r--  1 1001      www      3 Jun  1 14:24 data_bk_ver.txt
-rw-r--r--  1 1001      www      12 Jun 30 15:52 hello.py
-rwxr-xr-x  1 1001      www      946 Jun 23 10:32 index.html
-rw-r--r--  1 0         0       93988 Jun  1 10:00 tower.png
-rw-r--r--  1 1001      www     93840 Jun 10 12:53 tower2.png
-rw-r--r--  1 1001      www      7 Jun  1 11:28 wwwwpute

+CFTPS LIST: 0

AT+FSCD=F:/
+FSCD: F:/

OK

AT+FSLS
+FSLS: SUBDIRECTORIES:
usb
mifi

+FSLS: FILES:
data_bk_ver.txt
dnsmasq.conf
mobileap_cfg.xml

OK

Uploading file to "120.79.2.0"...
upload file from FTP...
AT+CFTRANRX="E:/hello.py",12 ERROR
AT+CFTRANRX="E:/hello.py",12 back:      AT+CFTRANRX="E:/hello.py",12
>
1314xal1314
OK

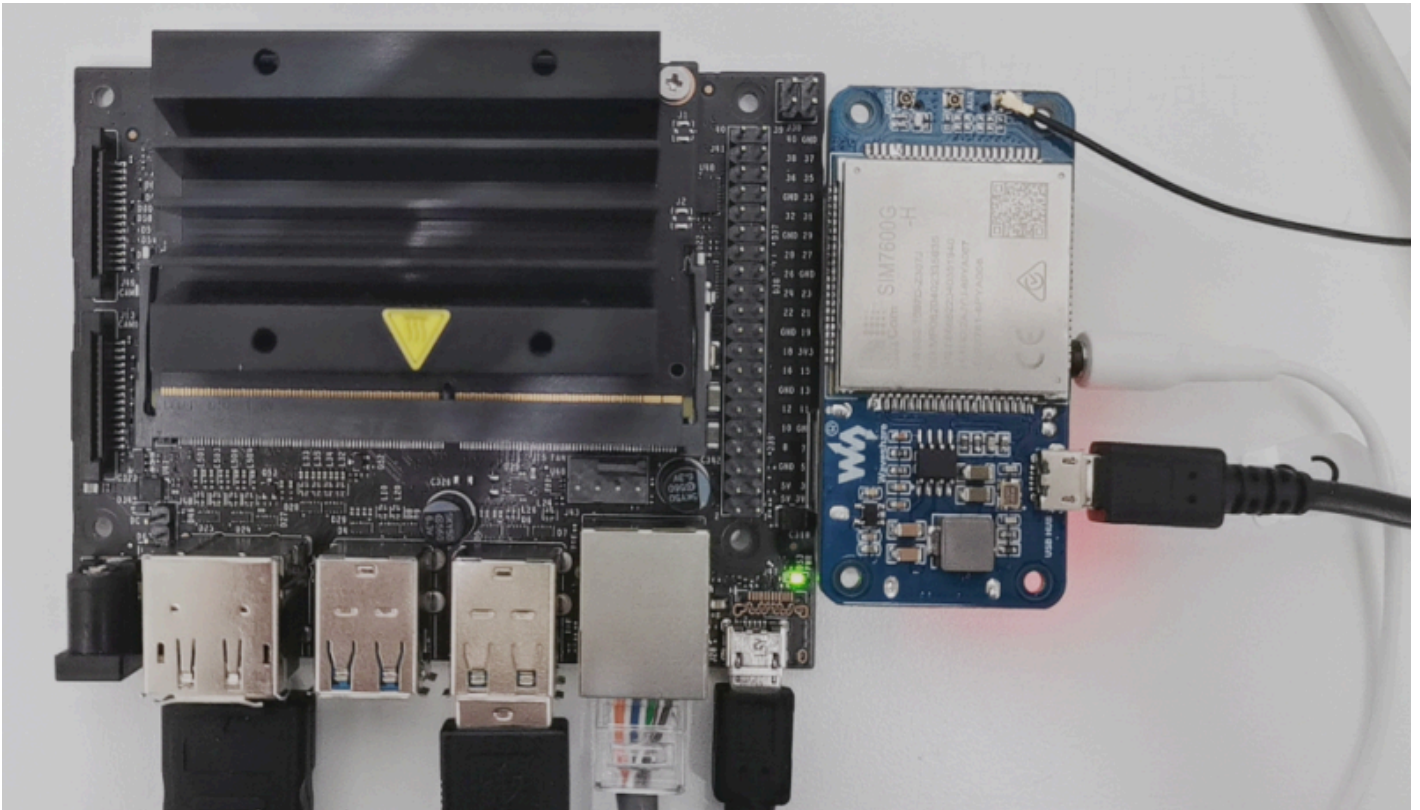
```

(/wiki/File:SIM7600G-H_4G_HAT_(B)_AT06.png)

Jetson Nano Demo

Hardware Connection

Connect the SIM7600X 4G HAT(B) to the USB port of Jetson Nano via Micro USB cable, as shown in the picture below:



(/wiki/File:SIM7600G-H_4G_HAT_Jetson_nano.png)

Minicom Serial Debugging for Jetson Nano

- Connect the SIM7600X 4G HAT(B) to Jetson Nano through the USB cable.
- Use SERIAL to log in to Jetson Nano (https://www.waveshare.com/wiki/Use_SERIAL_to_log_in_to_the_Jetson_Nano_terminal) terminal and install minicom:

```
sudo apt-get install minicom
```

- Run minicom for debugging, enter the following command in the terminal:

```
sudo minicom -D /dev/ttyUSB2 -b 115200
```

- Send AT command to test:

```
root@waveshare-desktop:/home/waveshare# sudo minicom -D /dev/ttyUSB2 -b 115200

Welcome to minicom 2.7.1

OPTIONS: I18n
Compiled on Aug 13 2017, 15:25:34.
Port /dev/ttyUSB2, 12:11:21

Press CTRL-A Z for help on special keys

AT

OK
AT+CSQ
+CSQ: 24,99

OK
```

(/wiki/File:SIM7600_G-H_B_Jetson.png)

Press Ctrl+A, then press X to exit the minicom.

More demos continue to be updated...

Resources

Document

- SIM7600G-H 4G HAT (B) V1 Schematic (https://files.waveshare.com/upload/f/fe/SIM7600X_4G_HAT_%28B%29.pdf)
- SIM7600G-H 4G HAT (B) V2 Schematic ([https://files.waveshare.com/upload/f/fe/SIM7600X_4G_HAT_\(B\)_V2.pdf](https://files.waveshare.com/upload/f/fe/SIM7600X_4G_HAT_(B)_V2.pdf))
- Size map (<https://www.waveshare.com/w/upload/c/c4/SIM7600-size.png>)

Demo

- SIM7600G-H 4G HAT (B) Demo (<https://files.waveshare.com/upload/4/4e/SIM7600G-H-4G-HAT-B-Demo.zip>)

Tools

- sscom (with sim7600 AT commands) (<https://files.waveshare.com/upload/9/95/SIM7600-AT-SSCOM-EN.7z>)
- SIMCom GPS (https://files.waveshare.com/upload/9/9f/SIMCom_GPS_DEMO.7z)
- NetAssist (<https://files.waveshare.com/upload/f/f6/NetAssist.7z>)

Related Application Cases

- How to turn Raspberry Pi Zero W into a 3G 4G Router (https://www.waveshare.com/wiki/How_to_set_up_Raspberry_Pi_Zero_W_as_a_3G_4G_router)

Software

- SIM7600 Driver (https://files.waveshare.com/upload/2/24/SIMCOM_Windows_USB_Drivers_V1.0.2.zip)

- CP2102 driver (https://files.waveshare.com/upload/6/62/CP210x_USB_TO_UART.zip)
- SIM7600 Serial Debug Assistant (<https://files.waveshare.com/upload/d/dc/SIM7600X-AT-SSCOM-EN.zip>)
- GPS Debug Tool (<https://files.waveshare.com/upload/c/cc/GPS-Software.7z>)
- TCP Test Tool (https://files.waveshare.com/upload/a/a6/TCP-Test_Tool.7z)
- Xshell (<https://files.waveshare.com/upload/b/b4/Xshell.7z>)
- VLC media player (<https://files.waveshare.com/upload/4/4f/VLC-media-player.7z>)
- Unicode Conversion Software (<https://files.waveshare.com/upload/b/b3/Unicode.7z>)

Application Note

- SIM7X00 Series_GPIO_Application Note
(/wiki/File:SIM7X00_Series_GPIO_Application_Note_V1.00.pdf)
- SIM7X00 Series_SAT_Application Note
(/wiki/File:SIM7X00_Series_SAT_Application_Note_V1.00.pdf)
- SIM7X00 Series_SMS_Application Note
(/wiki/File:SIM7X00_Series_SMS_Application_Note_V1.00.pdf)
- SMS Application notes (/wiki/File:AN_SMS_V1.01.pdf)
- SIM7X00 Series_TCPIP_Application Note
(/wiki/File:SIM7X00_Series_TCPIP_Application_Note_V1.00.pdf)
- SIM7600 Series_HTTP_AT Command Manual
(/wiki/File:SIM7500_SIM7600_SIM7800_Series_HTTP_AT_Command_Manual_V1.00.pdf)
- SIM7600 Series_SSL_AT Command Manual
(/wiki/File:SIM7500_SIM7600_SIM7800_Series_SSL_AT_Command_Manual_V1.00.pdf)
- SIM7X00 Series_UART_Application Note
(/wiki/File:SIM7X00_Series_UART_Application_Note_V1.00.pdf)
- SIM7X00 Series_GPS_Application Note
(/wiki/File:SIM7X00_Series_GPS_Application_Note_V1.00.pdf)
- SIM7600 Series_MQTT_ATC (/wiki/File:SIM7500_SIM7600_Series_MQTT_ATC_V1.01.pdf)
- SIM7X00_Audio_Application_Note (/wiki/File:SIM7X00_Audio_Application_Note_V1.00.pdf)
- SIM7600_Sleep_Mode_Application Note
(/wiki/File:SIM7100_SIM7500_SIM7600_Sleep_Mode_Application_Note_V1.01.pdf)
- SIM7600 Series_LBS_Application Note
(/wiki/File:SIM7100_SIM7500_SIM7600_Series_LBS_Application_Note_V1.00.pdf)
- SIM7600 Series_USB_AUDIO_Application Note
(/wiki/File:SIM7100_SIM7500_SIM7600_Series_USB_AUDIO_Application_Note_V1.03.pdf)
- SIM7600 Series_UIM_HOT_SWAP_Application Note
(/wiki/File:SIM7100_SIM7500_SIM7600_Series_UIM_HOT_SWAP_Application_Note_V1.01.pdf)
- SIM7600M22 Series_TTS_Application Note
(/wiki/File:SIM7100_SIM7600M22_Series_TTS_Application_Note_V1.02.pdf)
- SIM7600 Series_HSIC_LAN_Application_Note
(/wiki/File:SIM7600_Series_HSIC_LAN_Application_Note_V1.00.pdf)
- SIM7600 Series Hardware Design (/wiki/File:SIM7600_Series_Hardware_Design_V1.02.pdf)

- [SIM7600G-H Hardware Design \(/wiki/File:SIM7600G_SIM7600G-H_Hardware_Design_V1.00.pdf\)](/wiki/File:SIM7600G_SIM7600G-H_Hardware_Design_V1.00.pdf)

FAQ

Question: What could be the issue if the module connected via header pins isn't recognized by the Raspberry Pi Zero?

Answer:

Hardware Reasons:

- Connect the USB HUB interface to the computer via a USB cable to check if the computer recognizes the USB HUB. If it does, the module should be working fine. (Use a multimeter in beep mode) Check for poor connections in the USB connector or header pins.

Software Reasons: Additionally, if the USB port of the Raspberry Pi Zero is configured as a wired network card, it might not recognize the USB HUB.

- Add the following command in the config.txt file:

```
sudo nano /boot/config.txt  
dtoverlay=dwc2,dr_mode=host
```

Remove the following command from the cmdline.txt file:

```
sudo nano /boot/cmdline.txt  
module-load=dwc2,g_ether
```

Restart the Raspberry Pi:

```
sudo reboot
```

```
GNU nano 5.4 /boot/config.txt *
arm_boost=1

[all]
max_usb_current=1

hdmi_group=2
hdmi_mode=87
hdmi_cvt 1024 600 60 6 0 0 0
hdmi_drive=1

enable_uart=1

[HAT]
#dtoverlay=dwc2
dtoverlay=dwc2,dr_mode=host
```

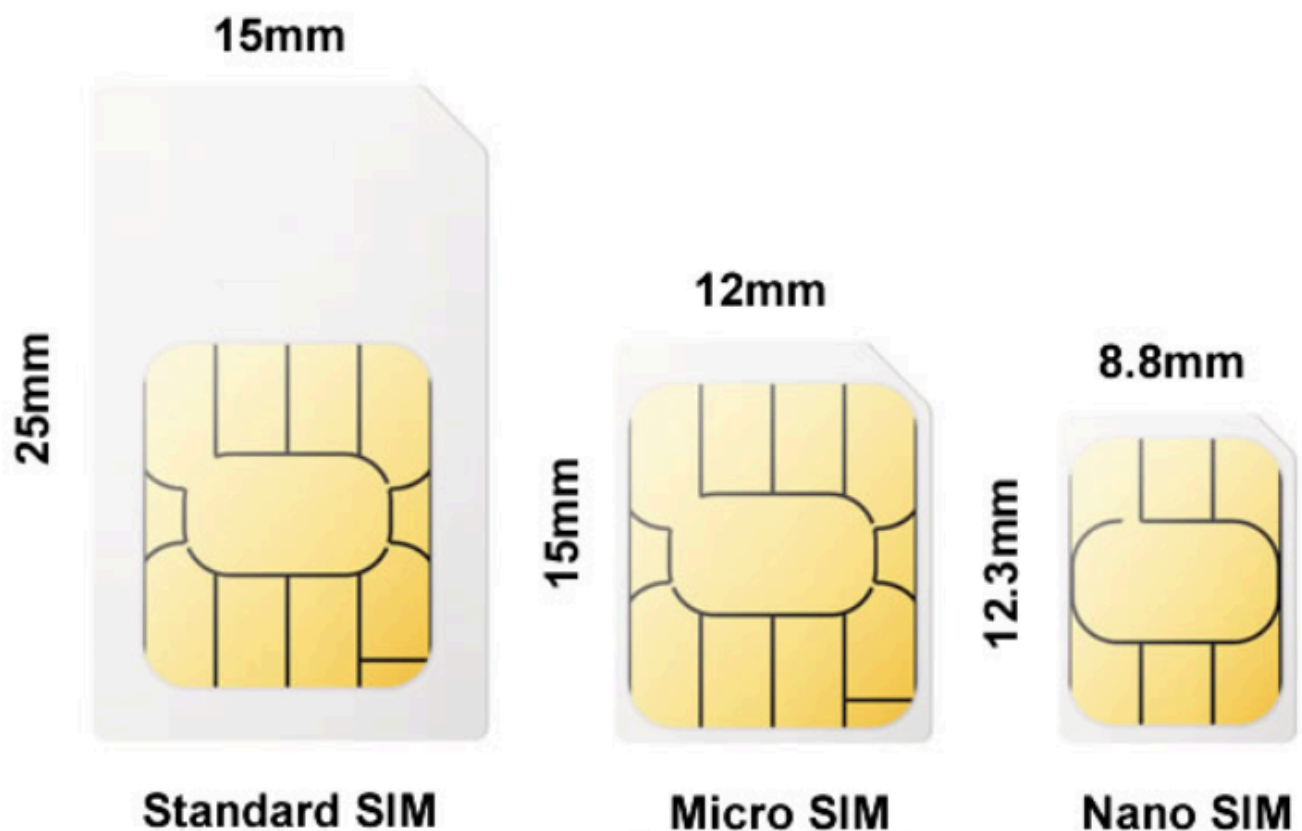
```
GNU nano 5.4 /boot/cmdline.txt *
console=tty1 root=PARTUUID=90ee35f4-02 rootfstype=ext4 fsck.repair=yes
rootwait module-load=dwc2,g_ether quiet splash plymouth.ignore-serial-consoles
```

(/wiki/File:Zero2W_SIM7600.png)

Question: Type of SIM card?

Answer:

- SIM7600X 4G HAT supports Standard SIM card.
- SIM7600X 4G Dongle supports Nano SIM card.



(/wiki/File:SIM_card_size.png)

Question: The NET light does not flash after the SIM7600X is turned on, what should I do if the network is abnormal?**Answer:**

In this case, it may be that you have not successfully connected to the network, you can follow the steps below to troubleshoot:

1. First check the hardware connection:

- Check if the MAIN antenna is well connected;
- Whether the connected SIM card can call and surf the Internet normally on mobile phones and other devices:
- It is recommended to replace the SIM cards of different operators. Under the comparison test, the network frequency bands and base station layouts supported by different operators are different. Changing the SIM card can also eliminate problems such as card arrears

2. After confirming that there is no problem with the hardware, the software can use these instructions:

- Check whether the sim card is in good contact: AT+CPIN?
- Check whether RF is turned on (flight mode is turned off): AT+CFUN?
- Check if the network mode setting is correct: AT+CNMP?
- Check the signal quality of the current environment: AT+CSQ
- Check the operator's access situation: AT+COPS?
- Check the connection status: AT+CPSI?
- Check the operator's access situation: AT+CGDCONT?
- Check the connection status: AT+SIMCOMATI
- Check whether it is successfully registered to the network: AT+CGREG?

```
AT+CPIN?  
+CPIN: READY
```

```
OK  
AT+CNMP?  
+CNMP: 2
```

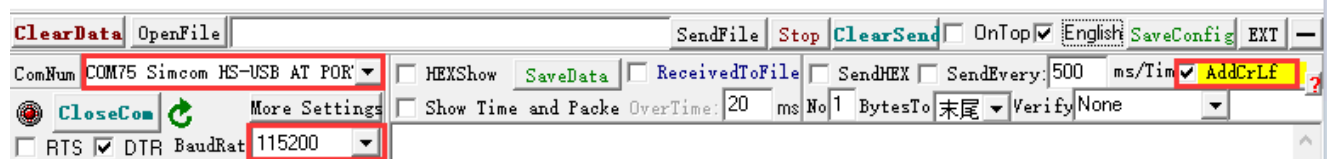
```
OK  
AT+CSQ  
+CSQ: 23, 99
```

```
OK  
AT+COPS?  
+COPS: 0, 0, "CHINA MOBILE CMCC", 7
```

```
OK  
AT+CPSI?  
+CPSI: LTE, Online, 460-00, 0x27B4, 205523841, 70, EUTRAN-BAND40, 38950, 5, 5, -  
101, -945, -645, 17
```

```
OK  
AT+CGREG?  
+CGREG: 0, 1
```

```
OK
```



(/wiki/File:SIM7600X_network_check.png)

Question: What are the models of SIM7600X series?

Answer:

SIM7600X mainly includes but is not limited to the following models:

模组型号	SIM7600A-H	SIM7600V-H	SIM7600E-H	SIM7600E-H1C(D)	SIM7600JC-H	SIM7600SA-H	SIM7600SA-H-MNSE	SIM7600NA-H	SIM7600G-H (R2)
网络制式	LTE CAT 4 150M/50M	LTE CAT 4 150M/50M	LTE CAT 4 150M/50M	LTE CAT 4 150M/50M	LTE CAT 4 150M/50M	LTE CAT 4 150M/50M	LTE CAT 4 150M/50M	LTE CAT 4 150M/50M	LTE CAT 4 150M/50M
尺寸大小	30mm*30mm*2.9mm	30mm*30mm*2.9mm	30mm*30mm*2.9mm	30mm*30mm*2.9mm	30mm*30mm*2.9mm	30mm*30mm*2.9mm	30mm*30mm*2.9mm	30mm*30mm*2.9mm	30mm*30mm*2.9mm
模组封装	87pin LCC	87pin LCC	87pin LCC	87pin LCC	87pin LCC	87pin LCC	87pin LCC	119pin LCC+LGA	119pin LCC+LGA
频段	LTE-FDD: B2/4/12 UMTS/HSPA+: B2/5	LTE-FDD: B2/4/5/13	LTE-TDD: B38/40/41 LTE-FDD: B1/3/5/7/8/20 UMTS/HSPA+: B1/5/8 GSM: 900/1800MHz	LTE-TDD: B38/40/41 LTE-FDD: B1/3/5/7/8/20 UMTS/HSPA+: B1/5/8 GSM: 900/1800MHz	LTE-FDD: B1/3/8/18/19/26	LTE-FDD: B1/2/3/4/5/7/8/28 LTE-TDD: B40 UMTS/HSPA+: B1/2/5/8 GSM: 850/900/1800/1900MHz	LTE-FDD: B1/2/3/4/5/7/8/28 LTE-TDD: B40 UMTS/HSPA+: B1/2/5/8 GSM: 850/900/1800/1900MHz	LTE-FDD: B2/4/5/12/13/14/25/26/6 6/71 LTE-TDD: B41	LTE-FDD: B1/2/3/4/5/7/8/12/13/18/19/20/25/26/28/66 LTE-TDD: B34/38/39/40/41 UMTS/HSPA+: B1/2/4/5/6/8/19 GSM: 850/900/1800/1900MHz
GNSS定位	可选	可选	可选	可选	可选	可选	可选	可选	可选
AT命令集	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
语音	可选	可选	可选	可选	可选	可选	可选	可选	可选
区域地区	北美地区 AT&T	北美地区 Verizon	欧洲以及中东	欧洲以及中东	日本	澳大利亚, 新西兰 台湾, 拉丁美洲	澳大利亚, 新西兰 台湾, 拉丁美洲	北美运营商 AT&T, Verizon, Sprint, T-mobile	全球
认证	FCC/PTCRB/IC/AT&T ROHS/REACH	FCC/GCF/Verizon/ ROHS/REACH	CE/NCC/GCF/ Vodafone*/ ROHS/REACH	CE/ ROHS*/REACH*	Telec/Uale/Softbank/ Docomo	CE/RCM/FCC/NCC/An atel/ ROHS/REACH/	CE/RCM/FCC/NCC/An atel/ ROHS/REACH/	FCC*/IC*/PTCRB*/GC F* AT&T*/Verizon*/Sprint*	CE/RCM*/FCC/IC/PTCRB*/ GCF* AT&T*/Verizon* * on going

(/wiki/File:SIM7600X_4G_DONGLE.PNG)

Question: What does the positioning information obtained by SIM7600X through AT+CGPSINFO represent?

Answer:

From left to right are ① Latitude, ② Longitude, ③ Date, ④ Time, ⑤ Altitude, ⑥ Speed and ⑦ Navigation Angle.

Question: What should I do if I can't receive the GPS signal and get the location information?

Answer:

Plug in the GPS antenna to the GNSS antenna socket, and place the receiver label face down in an open space (note that it cannot be tested in rainy weather). It takes about 1 minute to receive the positioning signal after power-on;

Question: After the GPS is turned on, why does the NMEA port not print the corresponding GPS information?

Answer:

Restart SIM7600E-H after sending the following commands:

```
AT+CGPSNMEAPORTCFG=3
AT+CGPSNMEA=197119
AT+CGPSINFOCFG=10, 31
```

Question: How to locate the base station for SIM7600X, what is the command?**Answer:**

Common commands for the LBS base station positioning function of SIM7600X are as follows:

```
AT+CLBS=? //View the range of parameters that can be set
AT+CNETSTART//Open the network; if it fails to open the network, you can use the command A
T+CNETSTOP to close and then open
AT+CLBS=1 //Get the current latitude and longitude
AT+CLBS=2 //Get the detailed address
```

Question: When sending AT+CPIN?, it returns ERROR?**Answer:**

This problem is generally caused by poor contact between the SIM card and the SIM card socket of the module.

Question: When sending an AT command, it can return OK, but the command sent is not displayed. Why?**Answer:**

The reason is that the echo is not turned on. SIM7600X can send the following command and press Enter, and the echo is successfully turned on after the display is OK.

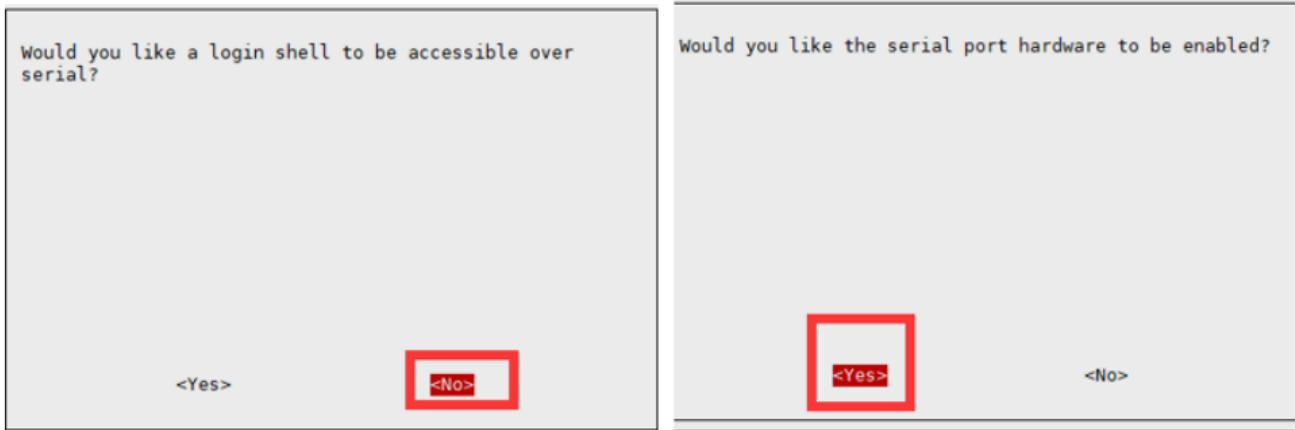
```
ATE1
```

Question: Why can't I use UART to control the module on the Raspberry Pi, I can't open the ttyS0 through minicom, and it prompts that there is no ttyS0?**Answer:**

Since the Raspberry Pi serial port is used for terminal debugging by default, if you need to use the serial port, you need to modify the Raspberry Pi settings. Execute the following command to enter the Raspberry Pi configuration:

```
sudo raspi-config
```

Select Interfacing Options ->Serial ->NO ->YES to disable serial debugging



(/wiki/File:Serial_port.png)

Open the /boot/config.txt file, and find the following configuration statement to enable the serial port, if not, add it at the end of the file:

```
enable_uart=1
```

Restart to take effect

Question: What should you do if you are registered to the network, dial up successfully and get an IP, but you cannot access the Internet and cannot ping through?

Answer:

It may be that the APN has not been obtained. Generally, the APN can be obtained automatically. In some areas (IoT card), it needs to be obtained manually. For example, it can be set by the following instructions:

```
AT+CGDCONT=1,"IP","Your_APN" //The APN of different operators is different, here the APN is
changed to the corresponding operator, for example:
China Mobile APN: CMNET; China Unicom APN: 3GNET; China Telecom APN: CTNET
```

AT+CGDCONT=1, "IP", "CMNET"
OK

(/wiki/File:SIM7600X_apn.png)

Question: How to make SIM7600X switch to IPV6 after dialing up the Internet?

Answer:

It can be set by the following commands:

```
AT+CGDCONT=1,"IPV6","APN" //Switch to IPV6, the APN of different operators is different, pay attention to distinguish the settings
AT+CGDCONT=1,"IP","APN" //Switch back to IPV4
```

Question: Why can I use an ordinary mobile phone SIM card that can make calls to access the Internet, but I cannot access the Internet using an IoT card (traffic card)?

Answer:

- No APN is set, configure APN as described above.
- After being banned, high-traffic (real-name IoT) cards are bound by chance cards and can only be used on one device (Ministry of Industry and Information Technology, Ministry of Public Security, must be issued to operators); The operator checks the status of the card and unlocks it.

Question: Some SIM card operators require the VOLTE function to make calls, how do I turn the VOLTE function on or off?

Answer:

The VOLTE function can be turned on with the following command:

```
at+voltesetting=1
at+cnv=/nv/item_files/modem/mmode/ue_usage_setting,1,01,1
```

The VOLTE function can be turned off with the following command:

```
at+voltesetting=0
at+cnv=/nv/item_files/modem/mmode/ue_usage_setting,0,01,1
```

Question: How to set the SMS center number? (SMS center number is a kind of short message server. The SMS sent by the mobile phone needs to be sent to the SMS center number first, and then forwarded to the other party's mobile

phone by the SMS center number, which is equivalent to a short message transfer station.)

Answer:

```
AT+CSCA="+8613800755500"
```

Command to add + Enter, return OK. Note: China Mobile's SMS service center number is +861380xxxx500, where xxxx is the long-distance telephone area code where you are located. The SMS center may be different from place to place. For details, you can query Baidu or call the customer service of China Mobile Unicom. This SMS center is Shenzhen (0755).

Question: SIM7600X fails to send SMS, prompts +CMS ERROR? or CME ERROR and other errors

Answer:

- Confirm that SIM7600X is registered to the network, and confirm that the SIM card can send and receive text messages normally on mobile phones and other devices;
- Set the correct SMS center number;
- Initialize SMS settings with the following commands:

```
AT+CSCS="IRA"  
AT+CSMP=17,167,0,0
```

Question: What is the function of the AUX antenna?

Answer:

The AUX antenna can increase the downlink rate: The AUX antenna is also a diversity antenna, which plays the role of receiving signals, improves the signal reception capability, and works with the MAIN antenna to increase the downlink rate.

Question: Why return NO CARRIER after NDIS dialing on the computer?

```
AT+QCRMICALL=1,1
```

```
[12:05:22.711] OUT → ◇ AT$QCRMCall=1, 1
[12:05:22.714] IN ← ◆ AT$QCRMCall=1, 1
NO CARRIER
```

(/wiki/File:SIM7600_dial_up_no.png)

Answer:

- Under normal circumstances, SIM7600X has been automatically dialed after receiving the Windows system, and there is no need to repeat dialing. Repeated dialing will return NO CARRIER
- If you still can't dial-up Internet access, please use the following command to change to Windows default dial-up Internet access mode

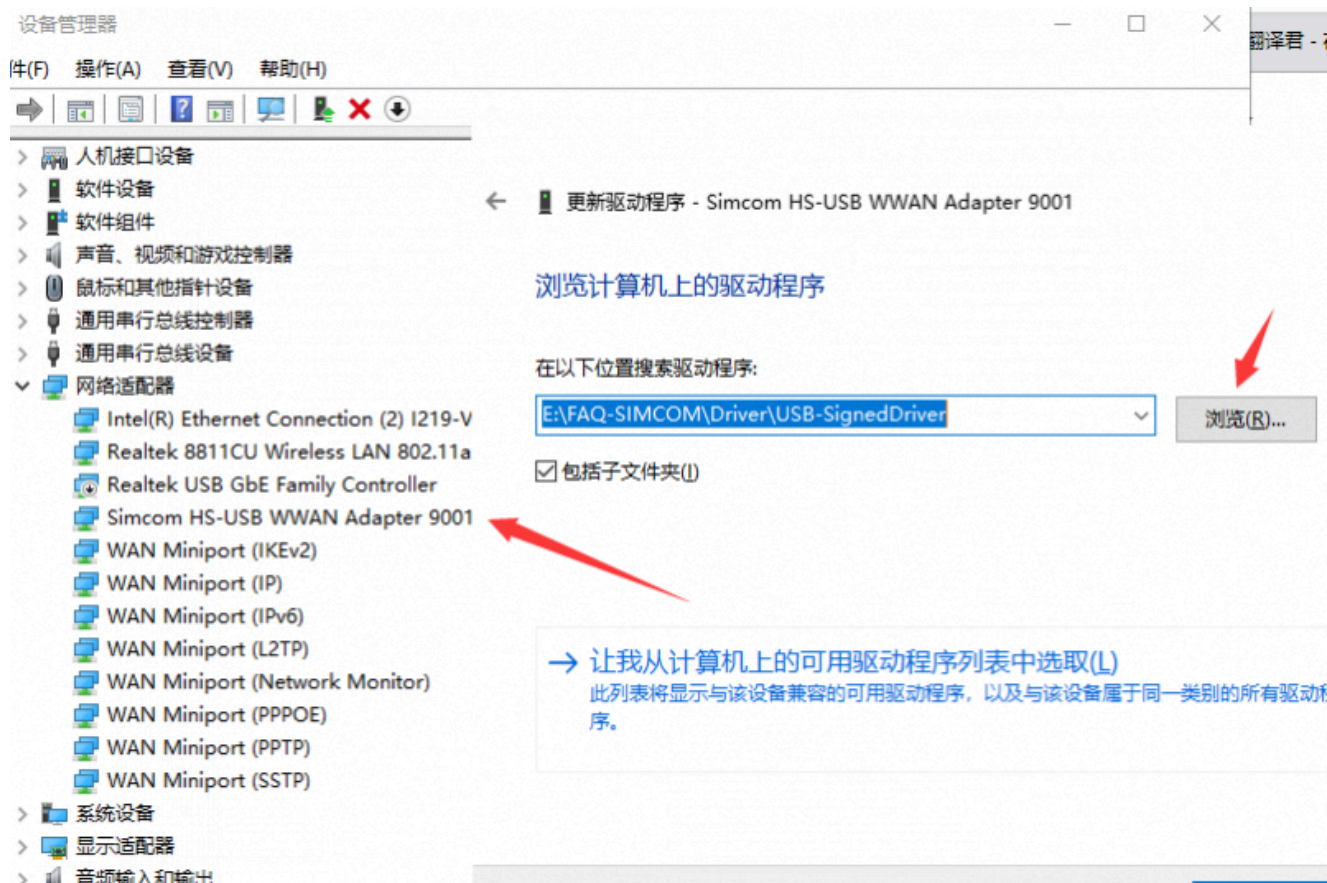
```
AT+CUSBPIDSWITCH=9001,1,1
```

- The display is turned off, and the mobile network is not enabled, you can ignore it and go online directly;



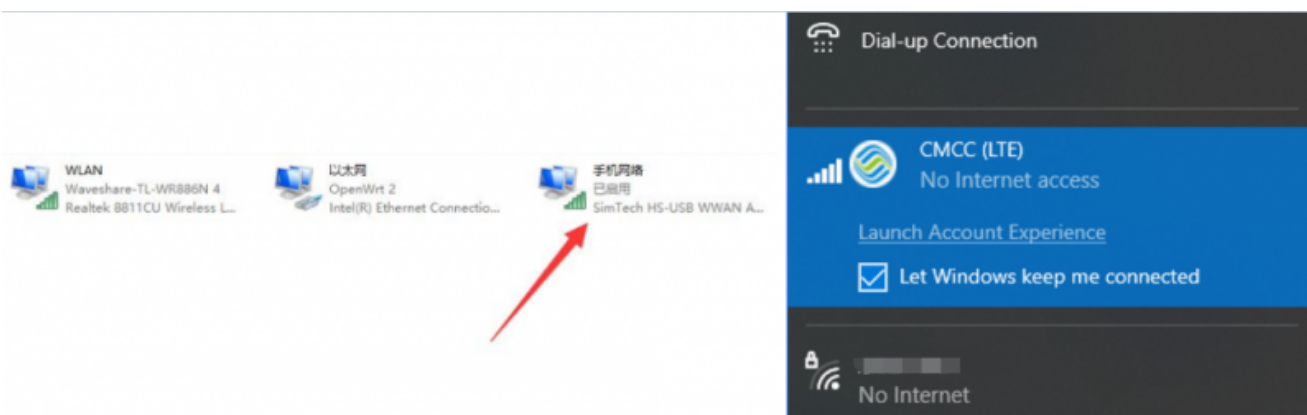
(/wiki/File:SIM7600X_rndis_dial-up_(1).png)

- You can also install the driver SIM7600X dial-up Driver (https://files.waveshare.com/upload/8/8a/SIM7600X_dial-up_Driver.zip) to update the network card.



(/wiki/File:SIM7600X_rndis_dial-up_(2).png)

- The network card display is enabled after installing the driver.



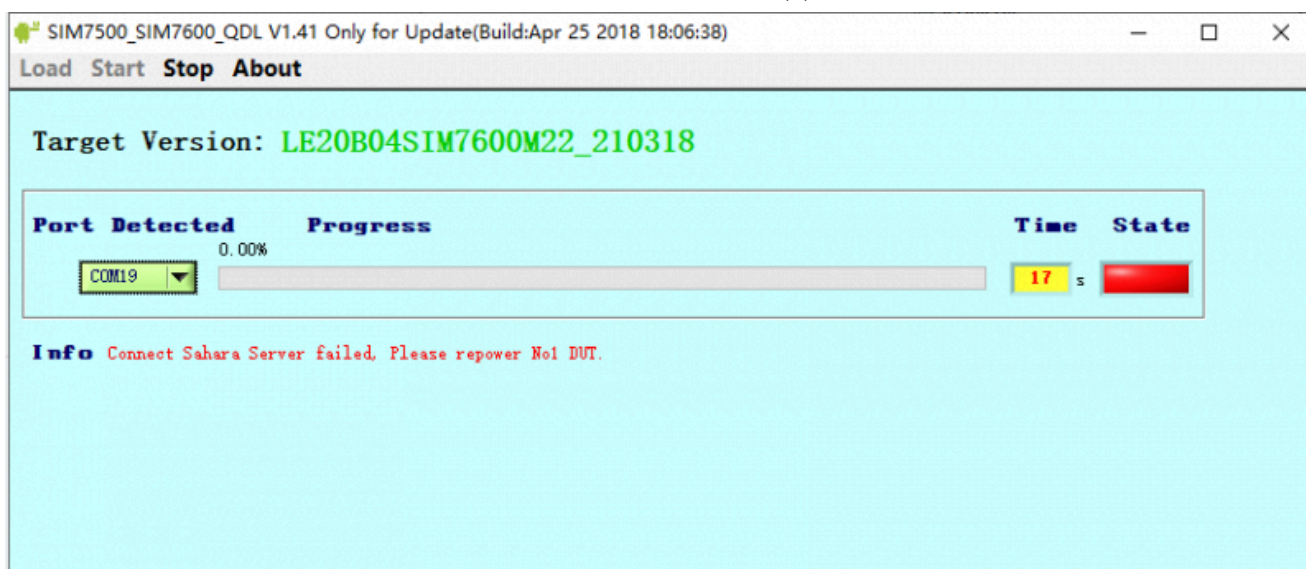
(/wiki/File:SIM7600X_rndis_dial-up_(3).png)

Question: Why is my dial-up internet speed very slow here?

Answer:

- Generally, the default configuration of SIM7600 is to automatically select the network standard, and it is likely to choose 2G Internet access; if you need to force the use of 4G mode, you need to enter the following AT command configuration:

AT+CNMP=38 //Fixed 4G LTE, if there is no local 4G coverage, you may not be able to register to the network



(/wiki/File:SIM7600X-FAQ_(6).png)

1. Pay attention to checking the device manager, the new device will be prompted during the upgrade process, and there will be no device driver during the first upgrade;
2. Pay attention to the USB cable. The speed of the USB cable is higher during the upgrade process. You need to choose a better-quality USB cable to avoid poor contact.
3. You need to run the upgrade tool with administrator privileges (SIM7500_SIM7600_QDL V1.41 only for Update)
4. Uninstall and reinstall the upgrade tool (SIM7500_SIM7600_QDL V1.41 only for Update)
5. For more operation details, please refer to this video:

<https://www.waveshare.net/wiki/SIM7600-Firmware-upgrade-Video>

(<https://www.waveshare.net/wiki/SIM7600-Firmware-upgrade-Video>)

Question: How many short messages can be stored in SIM7600CE 4G HAT?

Answer:

If the short message is stored in the SIM card, usually 50 is the upper limit. You can use the command: AT+CPMS? Make a query

Question: What is the working current when connected to the Internet?

Answer:

5V power supply BY USB, after the network is successfully connected, the current is generally in the range of 50~300mA, and the average is about 150mA (for reference only, depending on the network environment and network working status.)

Question: What are the antenna parameters of SIM7600CE 4G HAT?**Answer:**

*Frequency: 700m 800m 900m 1710-1920M 2010-2100M 2300-2400M 2500-2690M-5800MHZ

- Gain: 9dbi $\pm$ 0.7dbi

Question: What should I do if the SIM7600X module NDIS dialing fails and the driver cannot be installed?**Answer:**

*Make sure that your system kernel is above 5.4. Do not use sudo update to upgrade the Raspberry Pi to the latest version, otherwise, the kernel version will be upgraded to a version higher than the current firmware and will not be recognized.

- It is recommended to use the more convenient RNDIS dial
- Can burn the latest Raspberry Pi Raspbian system, reconfigure NDIS dial-up
- Or use the image that has been configured with the driver NDIS dial-up and start the Raspbian system image (the driver has been installed)

Question: I am using the WIN7 system, what should I do if the driver installation fails?**Answer:**

The new driver may not be compatible with some WIN7 systems, you can try the old driver:

SIM7600 old driver (https://www.waveshare.com/wiki/SIM7600X_old_Driver_install)

Question: SIM7600G-H 4G DONGLE How to install Android driver:**Answer:**

- Currently there is the driver source code of Android4-Android11, that is, the system version is Android4-Android11 only supported;
- Get the source code and SDK of the Android system of the target device (requires official image support), then add SIMCOM's Android system driver source code to the Android source code, and recompile (the compilation time varies from 1-10 hours, it is recommended to use a high-configuration PC operate);
- If the USB can be recognized, it means that the Android driver has been installed successfully. Set the module to 9011 mode or 9018 mode, and you can dial up to access the Internet;
- The steps to install the Android driver are cumbersome, and the operator is required to have certain Android system development experience and R&D capabilities.

File:SIM7600-Android_RIL.zip (/wiki/File:SIM7600-Android_RIL.zip)

Question: What is the difference between RNDIS, NDIS, PPP, and ECM?

Answer:

The IP obtained by different dialing methods is different and has different characteristics.

Please refer to the following table for details:

Interface	Dial-up Mode	PID	Module	Kernel Configuration	Device Node	Network Card/IP	Features
Physical Serial Port / USB-to-Serial	PPP	-	Qualcomm / ASR	CONFIG_PPP CONFIG_USB_SERIAL CONFIG_USB_SERIAL_WWAN CONFIG_USB_SERIAL_OPTION	ttyUSB	ppp0 Carrier IP	1. Dial via serial port, no network card driver required 2. Widely used, inherited from 2G/3G era 3. Low speed
USB Network Adapter	NDIS/QMI	9001	Qualcomm	CONFIG_USB_WDM CONFIG_USB_USBNET CONFIG_USB_NET_QMI_WWAN Kernel version 3.4 and above	cdc_wdm	wwan0/qmi mux0 Carrier IP	1. Requires network card driver provided by us 2. Supports QMAP, suitable for high-speed applications
USB Network Adapter	RNDIS	9011	Qualcomm / ASR	CONFIG_NETDEVICES CONFIG_USB_USBNET CONFIG_USB_USB_RNDIS_HOST	-	usb0 Qualcomm: Module LAN IP ASR: Carrier IP / Module LAN IP	1. System built-in driver 2. Internal automatic dial-up
	ECM	9018/ 9011	Qualcomm / ASR	CONFIG_NETDEVICES CONFIG_USB_USBNET CONFIG_USB_NET_CDCETHER	-	usb0 Qualcomm: Module LAN IP ASR: Carrier IP / Module LAN IP	1. System built-in driver 2. Internal automatic dial-up
	MBIM	9003/ 901E	Qualcomm	CONFIG_NETDEVICES CONFIG_USB_USBNET CONFIG_USB_NET_CDC_MBIM Kernel version 3.18 and above	cdc_wdm	wwan0/qmi mux0 Carrier IP	1. System built-in driver 2. Supports QMAP, suitable for high-speed applications

(/wiki/File:ECM-RNDIS-NDIS-7600.png)

Below is a detailed description of the various dials:

■ NDIS driver for Internet access (9001 mode)

This method must depend on the Linux system and is suitable for application scenarios that need to be developed using Linux network socket programming. After loading the driver into the kernel, connect the SIM7600 to the motherboard with a USB cable. After the SIM7600 is turned on, the wwan0 network can be recognized. port, you can access the Internet through this network port. The bottom layer of this method depends on the USB virtual serial port of SIM7600. This dial-up method can obtain the IP provided by the public operator, and the network speed is faster.

■ RNDIS (9011 mode)

RNDIS refers to Remote NDIS. The implementation of RNDIS based on USB is TCP/IP over USB, which is to run TCP/IP on the USB device, making the USB device look like a network card. This method only needs simple configuration, the motherboard will recognize the USB0 network card, and quickly obtain the USB0 network card and module or the operator's IP network access; RNDIS network speed is relatively fast, which is one of the most commonly used dial-up methods.

■ ECM (9018 mode)

These two are the "NDIS" standard under Linux. ECM is the abbreviation of the Ethernet Networking Control Model. ECM meets the requirements of the CDC on USB. The data call established through standard CDC-ECM is routed through the router, and the obtained IP address is a private IP such as 192.168; if the kernel supports this way, no additional driver is required. All data interacting with the module through the USB bus is constrained by relevant protocols and standards, and the module reaches the module through the USB hardware to complete the interaction with the Linux motherboard.

- PPP dial-up

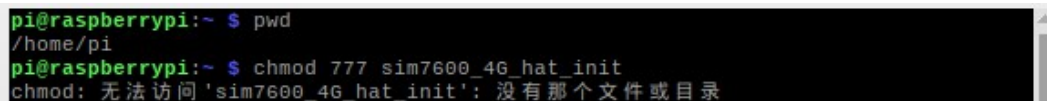
This method must depend on the Linux system and is suitable for application scenarios that need to be developed using Linux network socket programming. After configuring and running the relevant scripts, connect the SIM7600 with a USB cable. After the SIM7600 is powered on, dial up the pppd script to identify it. To the ppp0 network port, you can access the Internet through this network port and obtain the operator's IP. The bottom layer of this method depends on the USB virtual serial port of SIM7600.

- AT command uses encapsulated TCP, MQTT, HTTP(S)

This method is suitable for microprocessors with limited resources, such as MCU, or for application scenarios with a relatively small amount of data, such as uploading sensor data to servers, cloud platforms, etc. through http(s), MQTT. If the network application is not complicated and the amount of data is relatively small (such as transmitting sensor data to the server and receiving control commands from the server), the function can be quickly used by using AT commands.

- SIM7600X_RNDIS_Dial-Up_on_Windows (https://www.waveshare.com/wiki/SIM7600X_RNDIS_Dial-Up_on_Windows_10)

Question: When executing the `chmod 777 sim7600_4G_hat_init` command, an error is reported: "chmod: cannot access 'sim7600_4G_hat_init': No such file or directory"



```
pi@raspberrypi:~ $ pwd
/home/pi
pi@raspberrypi:~ $ chmod 777 sim7600_4G_hat_init
chmod: 无法访问 'sim7600_4G_hat_init': 没有那个文件或目录
```

([/wiki/File:777sim.png](#))

Answer:

Please confirm that there is a `sim7600_4G_hat_init` file in the current path.

```

pi@raspberrypi:~ $ cd SIM7600X/
pi@raspberrypi:~/SIM7600X $ ls
arduPi.cpp  AT      GPS      Makefile  sim7600_4G_hat_init  sim7x00.o
arduPi.h    bcm2835 main.cpp  PhoneCall sim7x00.cpp          SMS
arduPi.o    FTP      main.o    sim7600   sim7x00.h            TCP
pi@raspberrypi:~/SIM7600X $ ls sim7600_4G_hat_init
sim7600_4G_hat_init
pi@raspberrypi:~/SIM7600X $ chmod 777 sim7600_4G_hat_init
pi@raspberrypi:~/SIM7600X $

```

(/wiki/File:FAQ-

SIM7600X.png)

The general operation is: download the sample program, after decompression, rename the c folder under the Raspberry folder to SIM7600X, then copy the entire SIM7600X folder to the Raspberry Pi /home/pi directory, and enter the command line to /home/pi /SIM7600X directory, and then execute the chmod 777 sim7600_4G_hat_init command.

Question: When compiling the BCM2835 library, Makefile:327:recipe for target 'aclocal.m4' failed

```

config.status: executing depfiles commands
CDPATH="${ZSH_VERSION+.}:" && cd . && aclocal-1.13 -I m4
/bin/bash: aclocal-1.13: command not found
Makefile:327: recipe for target 'aclocal.m4' failed
make: *** [aclocal.m4] Error 127
root@raspberrypi:/home/pi/SIM7600X/bcm2835#

```

(/wiki/File:Autorreconf2.png)

Answer:

Execute: autoreconf -vfi, and then recompile, as shown in the following figure:

```
pi@raspberrypi:~/SIM7600X/bcm2835 $ sudo apt-get install autoconf
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  automake autotools-dev libsigsegv2 m4
Suggested packages:
  autoconf-archive gnu-standards autoconf-doc libtool gettext m4-doc
The following NEW packages will be installed:
  autoconf automake autotools-dev libsigsegv2 m4
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,358 kB of archives.
After this operation, 4,185 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://mirrors.nju.edu.cn/raspbian/raspbian stretch/main armhf libsigsegv2
  armhf 2.10-5 [28.4 kB]
Get:2 http://mirrors.nju.edu.cn/raspbian/raspbian stretch/main armhf m4 armhf 1.
  4.18-1 [185 kB]
Get:5 http://mirrors.nju.edu.cn/raspbian/raspbian stretch/main armhf automake al
  l 1:1.15-6 [733 kB]
Get:3 http://mirrors.nju.edu.cn/raspbian/raspbian stretch/main armhf autoconf al
  l 2.69-10 [338 kB]
Get:4 http://mirrors.nju.edu.cn/raspbian/raspbian stretch/main armhf autotools-d
  ev all 20161112.1 [73.4 kB]
Fetched 1,358 kB in 4s (290 kB/s)
```

```
pi@raspberrypi:~/SIM7600X/bcm2835 $ sudo autoreconf -vfi
autoreconf: Entering directory `.'
autoreconf: configure.ac: not using Gettext
autoreconf: running: aclocal --force -I m4
aclocal: warning: couldn't open directory 'm4': No such file or directory
autoreconf: configure.ac: tracing
autoreconf: configure.ac: not using Libtool
autoreconf: running: /usr/bin/autoconf --force
autoreconf: running: /usr/bin/autoheader --force
autoreconf: running: automake --add-missing --copy --force-missing
configure.ac:11: installing './compile'
autoreconf: Leaving directory `.'

```

(/wiki/File:Autorreconf.png)

Question: After the GPS is turned on, why does the NMEA port not print the corresponding GPS information?

Answer:

Restart SIM7600E-H after sending the following commands:

```
AT+CGPSNMEAPORTCFG=3
AT+CGPSNMEA=197119
AT+CGPSINFOCFG=10,31
AT+CGPSAUTO=1
```

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.

Working Time: 9 AM - 6 PM GMT+8
(Monday to Friday)

Submit Now (<https://service.waveshare.com/>)

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